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Practical week 10

1. **Investigate inode structures using commands such as ls-i, stat, and find. Create hard links and symbolic links and explain their impact on inode allocation.**

Code:



```
 UW PICO 5.09                          File: 102.c                          Modified

        return 1;
    }

    fprintf(fp, "Hello Linux Inodes!\n");
    fclose(fp);

    printf("=== Original File ===\n");
    show_inode("original.txt");

    /* Create hard link */
    if (link("original.txt", "hardlink.txt") == -1) {
        perror("Hard link");
        return 1;
    }

    printf("\n=== After Creating Hard Link ===\n");
    show_inode("original.txt");
    show_inode("hardlink.txt");

    /* Create symbolic link */
    if (symlink("original.txt", "symlink.txt") == -1) {
        perror("Symbolic link");
        return 1;
    }

    printf("\n=== After Creating Symbolic Link ===\n");
    show_inode("original.txt");
    show_inode("hardlink.txt");
    show_inode("symlink.txt");

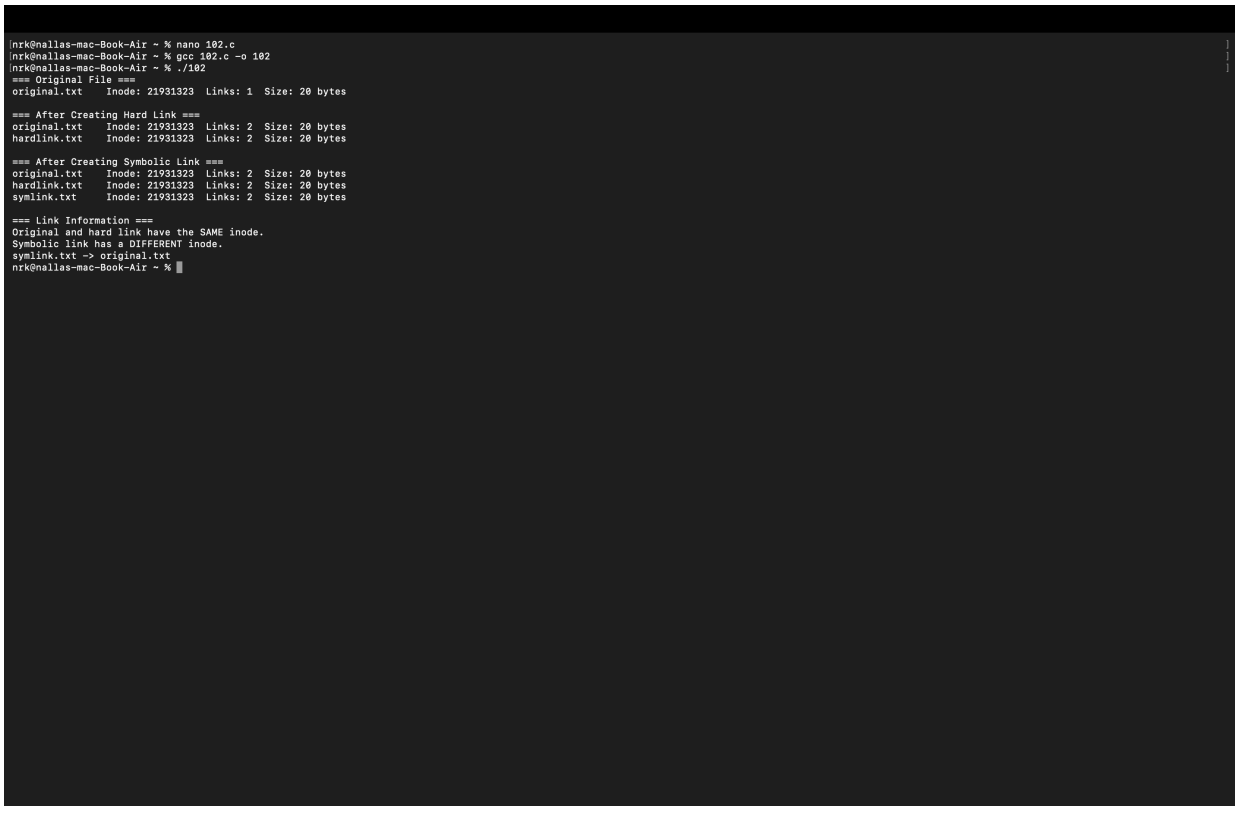
    printf("\n=== Link Information ===\n");
    printf("Original and hard link have the SAME inode.\n");
    printf("Symbolic link has a DIFFERENT inode.\n");

    /* Display symbolic link target */
    char buffer[100];
    ssize_t len = readlink("symlink.txt", buffer, sizeof(buffer) - 1);

    if (len != -1) {
        buffer[len] = '\0';
        printf("symlink.txt -> %s\n", buffer);
    }

    return 0;
}

^G Get Help  ^O WriteOut  ^R Read File  ^Y Prev Pg  ^K Cut Text  ^C Cur Pos
^X Exit      ^J Justify   ^W Where is   ^V Next Pg  ^U UnCut Text ^T To Spell
```



```
nrk@nallas-mac-Book-Air ~ % nano 102.c
nrk@nallas-mac-Book-Air ~ % gcc 102.c -o 102
nrk@nallas-mac-Book-Air ~ % ./102
=== Original File ===
original.txt    Inode: 21931323  Links: 1  Size: 20 bytes

=== After Creating Hard Link ===
original.txt    Inode: 21931323  Links: 2  Size: 20 bytes
hardlink.txt    Inode: 21931323  Links: 2  Size: 20 bytes

=== After Creating Symbolic Link ===
original.txt    Inode: 21931323  Links: 2  Size: 20 bytes
hardlink.txt    Inode: 21931323  Links: 2  Size: 20 bytes
symlink.txt     Inode: 21931323  Links: 2  Size: 20 bytes

=== Link Information ===
Original and hard link have the SAME inode.
Symbolic link has a DIFFERENT inode.
symlink.txt -> original.txt
nrk@nallas-mac-Book-Air ~ %
```

2.Design a program using mmap() to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity. same i have to do in ubuntu. this is practical 10.

Code:



Output:

